

The empirical recurrence for OEIS A296318: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A296318 records “Empirical recurrence of order 85”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 104$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 85, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 85 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A296318 is “Number of $n \times 6$ 0..1 arrays with each 1 horizontally, vertically or antidiagonally adjacent to 1 neighboring 1.”. It has offset 1 and begins

9, 132, 902, 8470, 77426, 676269, 6069953, 54182821, ...

The entry states:

Empirical recurrence of order 85 (see link above)

As of the “Last modified” line on the live entry (Dec 10 2017 11:30:57, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 104$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 104$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 208$ exact terms of the model returns order 85 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{85} c_i a(n-i),$$

c_1	1	c_{23}	−170172411	c_{45}	−19632194809	c_{67}	−4595507
c_2	34	c_{24}	−138246120	c_{46}	57069098674	c_{68}	15044368
c_3	254	c_{25}	548376965	c_{47}	−16644741108	c_{69}	1671757
c_4	541	c_{26}	363310138	c_{48}	−25230516411	c_{70}	−3035412
c_5	1147	c_{27}	−3722832580	c_{49}	34759355221	c_{71}	9043
c_6	−1356	c_{28}	4613086816	c_{50}	5648048196	c_{72}	478984
c_7	3203	c_{29}	14547300989	c_{51}	−29577153262	c_{73}	84081
c_8	−28096	c_{30}	−23498786767	c_{52}	7807264301	c_{74}	−82419
c_9	36458	c_{31}	−33010084429	c_{53}	20745674108	c_{75}	18613
c_{10}	−226969	c_{32}	58287129343	c_{54}	−9429113950	c_{76}	−2237
c_{11}	566326	c_{33}	40270296962	c_{55}	−10974952360	c_{77}	3925
c_{12}	−1750033	c_{34}	−86490524723	c_{56}	6537286062	c_{78}	−801
c_{13}	6453422	c_{35}	−12776345613	c_{57}	4261990865	c_{79}	822
c_{14}	−12737480	c_{36}	76349189039	c_{58}	−3805409336	c_{80}	−255
c_{15}	17922681	c_{37}	−45674551190	c_{59}	−1487739230	c_{81}	51
c_{16}	−22286052	c_{38}	−26300961979	c_{60}	1611274364	c_{82}	−11
c_{17}	44435546	c_{39}	96523771268	c_{61}	430396530	c_{83}	12
c_{18}	−79080791	c_{40}	−32684483322	c_{62}	−621121749	c_{84}	−1
c_{19}	−22286470	c_{41}	−100650521911	c_{63}	−119779838	c_{85}	1
c_{20}	85939463	c_{42}	73095558107	c_{64}	201376982		
c_{21}	303807920	c_{43}	63962995516	c_{65}	34583191		
c_{22}	−371846206	c_{44}	−81425595328	c_{66}	−57768396		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 85, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $85 + 4$ terms removed returns order 85 as well, so $\deg q_{\min} = 85 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $85 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 85 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A296318.
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